

Guideline Title: Trauma in Pregnancy

Guideline Number: GUID-3203-T017

Issue date: 7/1/2024	Replaces Dept. Policy:
Revision dates: 07/01/2026	Developed by: Air Care Team
Approved by: Dr. Danielle DiCesare, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

I. **PURPOSE:** The goal of therapy is to correct all life-threatening disorders in order to ensure adequate maternal tissue perfusion and adequate fetal blood flow. The general approach to trauma resuscitation in pregnant patients remains the same, with special considerations noted in this protocol

II. **DEFINITIONS:**

III. **PATHOPHYSIOLOGY**

- a. It is paramount that practitioners focus their interventions on behalf of the mother in order to save the fetus.
- b. Normal physiologic changes in pregnancy increase the risk of major trauma to both mother and fetus. Moderate blood loss or ventilatory problems can potentially lead to devastating consequences
- c. Decreased fetal heart rate and fetal movement are often indicators of maternal shock.
- d. Abdominal trauma may result in premature separation of the placenta (abruptio placentae), premature labor or abortion, rupture of the uterus, and fetal death.
- e. Special considerations may be required during maternal cardiac arrest. Postmortem Cesarean delivery is most effective within 5 minutes of maternal death. Viable pregnancy is an indication to *immediately* transport a cardiac arrest patient to the nearest emergency department. Provide early advance warning to the receiving facility in order to allow adequate time for trained staff members to assist in an emergent Cesarean delivery. *Postmortem Cesarean delivery may only be performed by credentialed physician providers in a hospital setting.*

IV. **DEPARTMENT GUIDELINE:**

- a. Assessment
 - i. Assess patient per *GUID-3203-M022 General Management of the Obstetric Patient and GUID-3203-T008 Multi-system Trauma*
 1. Obtain FHT as part of vital sign assessment
 2. If possible, ask patient to describe fetal movement
 - ii. During secondary assessment, careful attention should be paid to the abdomen such as rising fundal height, palpable fetal parts, obvious external signs of trauma, etc.
 - iii. Assess for signs and symptoms of maternal hypovolemia

Guideline Title: Trauma in Pregnancy

Guideline Number: GUID-3203-T017

- iv. Assess for potential chest injury. The mother's ventilatory status may already be compromised due to normal diaphragm elevation with pregnancy. Further compromise of ventilation may cause hypoxia that leads to shunting of blood from the uterus.
- v. Observe perineum and note amount and color of any vaginal bleeding.
- vi. Observe fetal heart rate by Doppler. If transferring from another facility, obtain copy of fetal monitoring strip and specifically note any decelerations.

b. Interventions

- i. All patients at risk for spinal injury require spinal motion restriction. Spinal board may be angled onto left side with aid of blankets or pillows placed under right side of board.
- ii. If patient is hypotensive, follow *GUID-3203-M011 - Hypotension*
- iii. For ongoing hemorrhage and hemorrhagic shock, administer blood products per *GUID-3203-PR002 Blood Product Administration* or *GUID-3203-PR025 Emergency Blood Product Administration*.
- iv. Monitor fetal heart rate by Doppler
- v. Immediately notify receiving facility if signs of fetal distress are present.
- vi. Reassess maternal perfusion and respiratory status if fetal distress is present.
- vii. Consider indwelling urinary catheter placement if a trauma transfer.
- viii. If patient suffers cardiac arrest, treat per the appropriate Patient Care Guideline.
 - 1. Consider air or ground transport to a facility capable of performing emergent Cesarean delivery of the fetus.

V. Special Considerations

a. Placental Abruption

- i. Consider placental abruption in any patient with a gravid uterus (greater than 20 weeks gestation) who has experienced blunt thoracoabdominal trauma. This is a clinical diagnosis that cannot be confirmed by sonogram. Patients with suspected placental abruption require an observation period and fetal monitoring at a facility with obstetric capabilities
 - 1. Signs and symptoms: abdominal pain (may range from mild to severe), vaginal bleeding, contractions, shock out of proportion to observed blood loss, evidence of fetal distress (abnormal FHT or fetal monitoring)
 - 2. If available, obtain fetal heart monitoring data (variability, decelerations), lab values (CBC, type and cross, fibrinogen), and imaging studies
 - 3. Interventions:
 - a. Keep the patient warm and provide supplemental oxygen, as needed.
 - b. Obtain fetal heart rate every 5 - 10 minutes
 - c. If patient has adequate spontaneous respirations, administer supplemental oxygen to maintain SpO2 > 93%.
 - d. Connect to cardiac monitor, NIBP, pulse oximeter and ETCO2 (when indicated to maintain ETCO2 35 - 45 mmHg).
 - e. Establish IV access x 2 large bore.

Guideline Title: Trauma in Pregnancy

Guideline Number: GUID-3203-T017

- f. Consider maintenance fluid NS/LR at 125 mL/hr IV.
- g. Volume resuscitate with IV fluid boluses of 0.9% NS/LR 500 mL IV as needed. Goal to maintain urine output above 30 mL/hour.
- h. Consider use of blood products for ongoing hemorrhage and hemodynamic instability per *GUID-3203-PR002 - Blood Product Administration* or *GUID-3203-PR025 - Emergency Blood Product Administration*.

b. Uterine Rupture

- i. May be a complication of pregnancy or due to blunt or penetrating trauma
- ii. Transport assessment and interventions are the same as placental abruption above

VI. **DOCUMENTATION:** EMS Charts

VII. **REFERENCES:**